Sensitive Data Accessing and Processing in Galaxy

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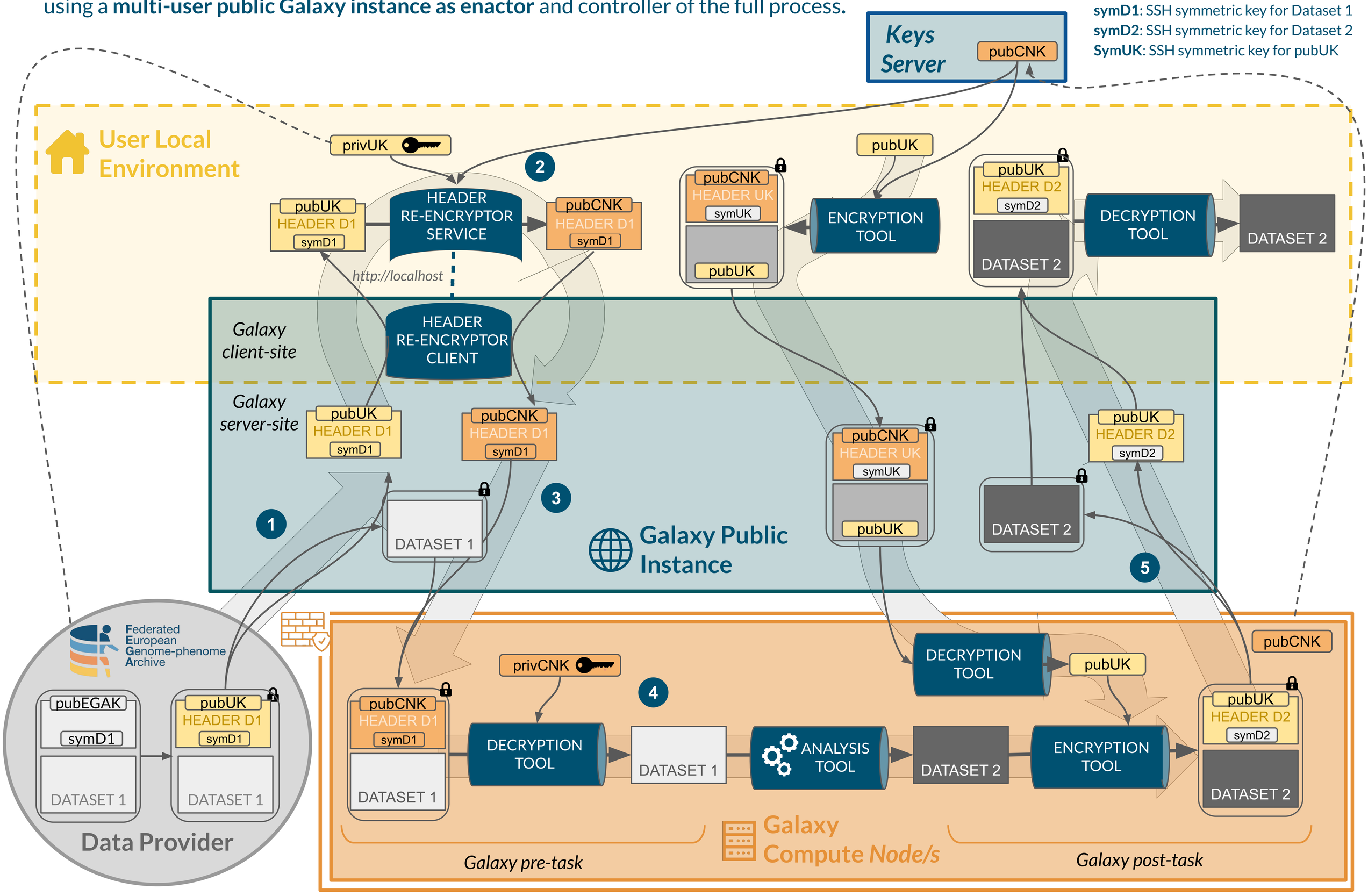
INTRODUCTION

In the context of the ELIXIR Implementation Study 'Strengthen Data Management in Galaxy', novel data handling strategies are being explored for enhancing the secure and scalable access and processing of sensitive datasets from Galaxy instances in a secure and scalable manner. To this end, we propose **an approach to access, transfer and process (F)EGA Crypt4GH encrypted datasets** into a secure environment using a **multi-user public Galaxy instance as enactor** and controller of the full process.

LEGEND
pubUK: User's SSH public key
privUK: User's SSH private key
pubCNK: Compute Node SSH public key
privCNK: Compute Node SSH private key
symD1: SSH symmetric key for Dataset 1
symD2: SSH symmetric key for Dataset 2
SymUK: SSH symmetric key for pubUK



1 Data import from (F)EGA to Galaxy server

Datasets are downloaded from (F)EGA as crypt4GH files encrypted using *User*'s key - i.e. only *User* is able to decrypt them. When stored in Galaxy, the header is split from the encrypted body, which contains the sensitive dataset.

A new Galaxy data import tool is being developed. The NO-FEGA REST API is being queried using the OIDC token issued by ELIXIR Life Science AAI when login in Galaxy.

2 Crypt4GH header re-encryption

To enable an eventual data decryption in the Galaxy compute backend, the original header is send to User's premises, where it is decrypted and encrypted again, this time using the *Compute Node* key.

A re-encryption REST service is being developed. To avoid User's private key being exported, the service runs in the local environment of the User. An extended version of the Galaxy client-site web application is able to reach the endpoint and carry out the transformation.

Short-lived keys of the compute Node/s are to be publicly exposed. An automatic system for renewing and provisioning them is required.

3 Galaxy Server as middleware

In the Galaxy instance, data is kept always encrypted, under the key of either the *User* or the *Compute Node*. In this way, Galaxy administrator never access *User*'s data.

4 Running an analysis tool: input decryption

Before running an analysis tool or workflow, input data is decrypted on the compute node, directly before being processed. Analysis tools remain unaltered.

An enhanced Galaxy instance could extend the common template launching jobs to Galaxy runners. It would include the instructions on how to encrypt/decrypt data in the job destination.

5 Running an analysis tool: output encryption

Output is encrypted using *User's* key, which has been sent from the Galaxy server as an extra crypt4GH file. Results are downloaded and locally decrypted.

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